

Workshop: Partial Differentiation

UEA LEARNING ENHANCEMENT TEAM MATHEMATICS & STATISTICS

1. For each of the functions below find

$$\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, \quad \frac{\partial^2 z}{\partial x^2}, \frac{\partial^2 z}{\partial y^2}, \frac{\partial^2 z}{\partial x \partial y}, \frac{\partial^2 z}{\partial y \partial x}$$

a. $z = 5x + 2y$

$$\frac{\partial z}{\partial x} = z_x = 5$$

$$\frac{\partial z}{\partial y} = z_y = 2$$

$$\frac{\partial^2 z}{\partial x^2} = 0, \quad \frac{\partial^2 z}{\partial y^2} = 0, \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = 0$$

b. $z = 5x^3 + 3y^2 + 4$

$$\frac{\partial z}{\partial x} = 15x^2$$

$$\frac{\partial z}{\partial y} = 6y$$

$$\frac{\partial^2 z}{\partial x^2} = 30x, \quad \frac{\partial^2 z}{\partial y^2} = 6, \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = 0$$

c. $z = 8xy$

$$\frac{\partial z}{\partial x} = 8y$$

$$\frac{\partial z}{\partial y} = 8x$$

$$\frac{\partial^2 z}{\partial x^2} = 0, \quad \frac{\partial^2 z}{\partial y^2} = 0, \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = 8$$

d. $z = 2xy^{0.5}$

$$\begin{aligned}\frac{\partial z}{\partial x} &= z_x = 2y^{0.5} \\ \frac{\partial z}{\partial y} &= z_y = xy^{-0.5} \\ \frac{\partial^2 z}{\partial x^2} &= 0, \quad \frac{\partial^2 z}{\partial y^2} = -0.5xy^{-1.5}, \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = y^{-0.5}\end{aligned}$$

e. $z = e^{2x} + 4e^y$

$$\begin{aligned}\frac{\partial z}{\partial x} &= 2e^{2x} \\ \frac{\partial z}{\partial y} &= 4e^y \\ \frac{\partial^2 z}{\partial x^2} &= 4e^{2x}, \quad \frac{\partial^2 z}{\partial y^2} = 4e^y, \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = 0\end{aligned}$$

f. $z = x^2 + 2xy + y^2$

$$\begin{aligned}\frac{\partial z}{\partial x} &= 2x + 2y \\ \frac{\partial z}{\partial y} &= 2x + 2y \\ \frac{\partial^2 z}{\partial x^2} &= 2, \quad \frac{\partial^2 z}{\partial y^2} = 2, \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = 2\end{aligned}$$

g. $z = 10x^2y^3 - 2x - 4y$

$$\begin{aligned}\frac{\partial z}{\partial x} &= 20xy^3 - 2 \\ \frac{\partial z}{\partial y} &= 30x^2y^2 - 4 \\ \frac{\partial^2 z}{\partial x^2} &= 20y^3, \quad \frac{\partial^2 z}{\partial y^2} = 60x^2y, \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = 60xy^2\end{aligned}$$

h. $z = 5x^2 + 2x + 5y^3 - 2y^2$

$$\begin{aligned}\frac{\partial z}{\partial x} &= 10x + 2 \\ \frac{\partial z}{\partial y} &= 15y^2 - 4y \\ \frac{\partial^2 z}{\partial x^2} &= 10\end{aligned}$$

$$\frac{\partial^2 z}{\partial y^2} = 30y - 4$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = 0$$

i. $z = 10x^4y^3$

$$\frac{\partial z}{\partial x} = 40x^3y^3$$

$$\frac{\partial z}{\partial y} = 30x^4y^2$$

$$\frac{\partial^2 z}{\partial x^2} = 120x^2y^3$$

$$\frac{\partial^2 z}{\partial y^2} = 60x^4y$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = 120x^3y^2$$

j. $z = 2e^{5x} - 4x^2y^2 + \ln 3xy = 2e^{5x} - 4x^2y^2 + \ln 3 + \ln x + \ln y$

$$\frac{\partial z}{\partial x} = 10e^{5x} - 8xy^2 + \frac{1}{x}$$

$$\frac{\partial z}{\partial y} = -8x^2y + \frac{1}{y}$$

$$\frac{\partial^2 z}{\partial x^2} = 50e^{5x} - 8y^2 - \frac{1}{x^2}$$

$$\frac{\partial^2 z}{\partial y^2} = 8x^2 - \frac{1}{y^2}$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = -16xy$$

k. $z = 2x^2e^{3y}$

$$\frac{\partial z}{\partial x} = 4xe^{3y}$$

$$\frac{\partial z}{\partial y} = 6x^2e^{3y}$$

$$\frac{\partial^2 z}{\partial x^2} = 4e^{3y}, \quad \frac{\partial^2 z}{\partial y^2} = 18x^2e^{3y}, \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = 12xe^{3y}$$

l. $z = x^3 \ln y$

$$\begin{aligned}\frac{\partial z}{\partial x} &= 3x^2 \ln y \\ \frac{\partial z}{\partial y} &= x^3 \frac{1}{y} = \frac{x^3}{y} \\ \frac{\partial^2 z}{\partial x^2} &= 6x \ln y, \quad \frac{\partial^2 z}{\partial y^2} = -\frac{x^3}{y^2}, \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = \frac{3x^2}{y}\end{aligned}$$

m. $z = y(5 - 2x) = 5y - 2xy$

$$\begin{aligned}\frac{\partial z}{\partial x} &= -2y \\ \frac{\partial z}{\partial y} &= 5 - 2x \\ \frac{\partial^2 z}{\partial x^2} &= 0, \quad \frac{\partial^2 z}{\partial y^2} = 0, \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = -2\end{aligned}$$

n. $z = y(5 - 2x)^2$

$$\begin{aligned}\frac{\partial z}{\partial x} &= -4y(5 - 2x) \\ \frac{\partial z}{\partial y} &= (5 - 2x)^2 \\ \frac{\partial^2 z}{\partial x^2} &= 8y, \quad \frac{\partial^2 z}{\partial y^2} = 0, \quad \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial^2 z}{\partial y \partial x} = -4(5 - 2x)\end{aligned}$$

First Order Conditions

2. For each of the functions $f(x, y)$ given find:

i. f_x and f_y

ii. Find the value of x, y , and $f(x, y)$ where both $f_x = 0$ and $f_y = 0$

a. $f(x, y) = x^2 + y^2$

$$f_x = 2x = 0$$

$$f_y = 2y = 0$$

$$x = 0, y = 0, f(x, y) = 0$$

b. $f(x, y) = x^2 - 2x - y^2 - 4y - 3$

$$f_x = 2x - 2 = 0$$

$$f_y = -2y - 4 = 0$$

$$x = 1, y = -2, f(x, y) = 0$$

c. $f(x, y) = x^2y - 2xy^2 + 3xy + 4$

$$f_x = 2xy - 2y^2 + 3y = 0$$

$$f_y = x^2 - 4xy + 3x = 0$$

Factorising:

$$f_x = y(2x - 2y + 3) = 0$$

$$f_y = x(x - 4y + 3) = 0$$

$$x = 0, y = 0, f(x, y) = 4$$

$$x = 0, 2x - 2y + 3 = 0 \Rightarrow 2y = 3 \Rightarrow y = 1.5$$

$$x = 0, y = 1.5, f(x, y) = 4$$

$$y = 0, x - 4y + 3 = 0 \Rightarrow x = -3$$

$$x = -3, y = 0, f(x, y) = 4$$